Anand Vardhan

B.Tech - 3rd Year
Mechatronics Engineering
Indian Institute of Technology, Bhilai

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Education

- **INDIAN INSTITUTE OF TECHNOLOGY, Bhilai** *2022-2026*

B.Tech - Mechatronics Engineering (Cgpa: 7.61(Sem-VI))

Experience

- **Bariflo Cybernetics Pvt. Ltd.** *May 2025 - Present*

Mechatronics Intern (GET) Bhubaneswar

– I am working on the Variable Buoyancy System(VBS) and dynamics for the development of a 3 dof Underwater Drone to be used for cleaning and inspection works in Fisheries and Aquaculture farms

– **Tools & technologies used**: SolidWorks/Fusion360, STM32, Sensor Integration

Projects

- **INTER-IIT 13.0- Centralized-Intelligence for Dynamic Swarm Navigation** Github repo

Designed and implemented a swarm navigation solution for autonomous robots to operate in dynamic environments without GPS.

– Developed a MongoDB-based system for real-time data logging and map updates, integrating ROS2 for swarm communication and obstacle avoidance.

– Achieved the 17th rank in the Kalyani BharatForge problem statement at Inter-IIT Tech Meet 13.0 hosted by IIT Bombay.

- **JACK – The Robot Dog** Github repo

A Quadrupedal robot for robust and agile movement in rough terrains by implementing Inverse Kinematics.

– Each foot has 3 DOF to ensure versatile and lifelike motion. Initially developed with Arduino, the project further leveraged ROS2 capabilities for simulation, calibration, and enhanced control algorithms for a trot gait-based stabilized walking.

- **Voice Command Guided Mobile Robot**

A Mobile Robot Control system for deployment in uninhabitable/toxic regions using voice command by ESP-NOW protocol

– Tested various ESP32 communication modes and selected ESP-Now protocol for sending and receiving command data over 120m from a remote.

– Enables exploration of contaminated areas, nuclear zones, and ecological data collection without direct human intervention.

Skills

Programming Languages/Scripts: Python, C / C++, Bash script, Java, HTML, CSS, JavaScript
Frameworks & Tools: ROS2, Gazebo, PID Control, Sensors, MicroControllers(Arduino, ESP, STM32), Internet of Things, MQTT, Socket Programming, Docker
Design & Simulation: Autodesk Fusion 360, PCB Design (KiCAD), MATLAB
Version Control: Git, Github
Some Coursework: Machine Learning, Embedded Systems for Mechanical Controls, Design & Analysis of Robotic Systems, Control Systems, Digital Controls, Mechanisms & Controls Labs, Advanced Sensing Techniques
Other Skills: Figma, Organizational Skills

Certifications

- IoT, Robotics and Embedded Systems | Coincent.ai *2023*
- ROS2 Nav2 [Navigation 2 Stack] – with SLAM and Navigation | Udemy *2025*

Positions of Responsibility

- **Co-Founder,** Entrepreneurship-Cell, IIT Bhilai *07/2024 - 05/2024*
- **Internship Coordinator,** Centre for Career Planning and Services, IIT Bhilai *04/2024 - 04/2025*
- **Public Relations Head,** Outreach Committee, IIT Bhilai *04/2024 - 10/2024*
- **Core Member,** The Epsilon Club(Robotics Club), IIT Bhilai *03/2024 - 04/2025*
- **Core Member(Aeromodelling Division),** Space Exploration Society, IIT Bhilai *04/2023 - 03/2024*
- **Volunteer,** Student Mentorship Programme, IIT Bhilai *04/2023 - 03/2024*
- **Graphic Designer,** Sports Council 23-24, IIT Bhilai *04/2023 - 02/2024*